

TAIF, Saudi Arabia

WMO: 410360

Lat: 21.483N

Lon: 40.544E

Elev: 4848

StdP: 12.30

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.4	47.0	10.7	11.4	81.3	16.6	15.2	75.4	23.9	69.3	22.3	69.8	3.9	180	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.8	98.9	62.0	97.7	62.0	96.6	61.9	71.2	86.9	69.2	86.1	67.7	86.0	10.3	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	116.2	81.4	64.1	107.6	78.8	62.3	100.7	76.5	38.6	86.1	36.6	86.4	35.1	85.9	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	21.1	19.1	DB	38.2	102.4	2.3	2.8	36.5	104.4	35.2	106.1	33.9	107.7	32.1	109.8	(4)
(5)			WB	32.2	75.9	1.9	5.1	30.8	79.6	29.6	82.6	28.5	85.5	27.1	89.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	74.4	61.0	64.7	68.6	73.9	79.8	85.3	85.4	86.1	82.7	74.6	67.4	62.6	(6)
(7)		DBStd	9.70	4.43	4.35	4.53	3.75	3.67	2.83	3.35	2.63	2.80	3.16	3.41	3.67	(7)
(8)		HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	310	135	53	20	2	0	0	0	0	0	0	14	86	(9)
(10)		CDD50	8910	341	411	578	717	925	1059	1098	1120	982	764	523	392	(10)
(11)		CDD65	3745	10	44	133	269	460	609	633	655	532	299	88	13	(11)
(12)		CDH74	46003	151	474	1250	2583	5342	8353	8340	8623	6698	3175	823	191	(12)
(13)	CDH80	22723	10	62	288	954	2682	4702	4467	4657	3524	1244	126	7	(13)	
(14)	Wind	WSAvg	8.4	6.9	8.0	9.0	8.6	7.8	9.9	12.5	11.5	7.8	6.5	6.3	6.4	(14)
(15)	Precipitation	PrecAvg	5.9	0.4	0.0	0.4	1.1	1.2	0.1	0.1	0.6	0.4	0.6	0.8	0.2	(15)
(16)		PrecMax	11.8	1.5	0.6	2.1	4.4	5.6	0.8	0.4	5.3	2.2	4.6	9.3	1.3	(16)
(17)		PrecMin	2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.5	0.4	0.1	0.6	1.1	1.4	0.2	0.1	1.1	0.6	1.0	1.8	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.0	84.5	88.2	91.7	97.2	100.5	100.7	100.5	98.7	91.7	85.8	80.7	(19)
MCWB			57.0	58.0	59.1	60.0	61.5	62.0	61.8	62.2	62.2	59.3	58.9	57.3	(20)	
2%		DB	77.4	82.1	85.8	89.7	95.4	98.7	98.7	98.7	96.7	89.8	82.8	78.3	(21)	
		MCWB	56.4	58.1	58.6	60.9	61.2	61.4	61.9	62.8	61.3	58.8	59.4	57.2	(22)	
5%		DB	75.2	79.2	83.0	87.8	93.6	97.1	96.8	97.0	95.0	88.1	81.0	76.0	(23)	
		MCWB	56.8	58.2	58.8	61.4	60.9	61.1	62.1	63.3	61.1	58.3	58.8	57.6	(24)	
10%		DB	73.1	77.0	80.8	85.8	91.6	95.4	94.9	95.2	93.3	86.4	78.9	73.8	(25)	
		MCWB	56.4	57.9	58.8	61.0	61.2	60.9	61.7	63.5	60.9	58.2	58.5	57.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.6	64.4	66.2	68.7	71.2	71.6	72.4	75.3	71.6	70.2	68.7	63.2	(27)
MCD B			73.4	75.9	78.4	82.9	86.3	90.0	89.0	89.9	85.6	79.0	76.9	73.5	(28)	
2%		WB	60.9	62.3	64.2	66.3	67.5	68.1	69.1	71.3	68.9	66.4	64.6	61.6	(29)	
		MCD B	71.6	74.6	77.0	80.6	84.6	89.7	88.0	87.9	83.7	75.3	74.1	72.3	(30)	
5%		WB	59.5	61.0	62.6	64.8	66.0	66.1	67.2	69.2	67.1	64.7	63.0	60.5	(31)	
		MCD B	70.0	73.7	76.5	79.5	83.8	89.1	88.3	87.5	83.9	75.5	73.4	71.3	(32)	
10%		WB	58.3	59.8	61.2	63.5	64.6	64.2	65.1	67.5	65.3	63.3	61.8	59.4	(33)	
		MCD B	68.9	72.6	75.6	78.8	82.8	88.9	88.2	87.8	84.3	75.9	72.9	70.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.6	24.5	24.4	23.9	23.9	21.9	19.0	19.8	23.1	24.6	23.5	23.4	(35)
MCDBR			26.1	26.3	26.1	25.5	25.1	22.8	20.5	21.5	23.6	25.5	26.0	25.7	(36)	
MCWBR			12.6	12.5	11.5	10.0	9.7	9.9	10.0	10.4	10.6	11.3	11.4	12.0	(37)	
5% WB		MCDBR	23.1	24.7	23.5	23.5	23.2	21.9	19.2	20.5	23.3	24.5	23.3	22.9	(38)	
		MCWBR	12.5	12.7	11.6	10.5	9.9	11.2	11.3	11.7	12.1	12.2	10.9	11.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.344	0.377	0.411	0.457	0.505	0.517	0.540	0.565	0.510	0.428	0.380	0.352	(40)
taud		2.310	2.187	2.070	1.963	1.828	1.786	1.764	1.708	1.802	2.009	2.180	2.276	(41)		
Ebn at Noon		293	289	283	271	256	250	245	239	251	268	277	285	(42)		
Edh at Noon		37	45	53	60	68	70	72	76	68	53	43	38	(43)		
(44)	All-Sky Solar Radiation	RadAvg		1530	1816	2091	2217	2371	2520	2395	2235	2125	1937	1603	1456	(44)
(45)		RadStd		82	69	85	101	107	66	60	45	56	60	72	67	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.63	N/A	N/A	N/A	N/A	N/A	-63	+277	+229
(47) Regional (3 neighbors)		+0.86	+1.26	-2.23	N/A	N/A	N/A	N/A	+338	+310

Nomenclature: See separate page